# HH-P-117
**NOVEMBER 5, 1940**

## FEDERAL SPECIFICATION

# PACKING; JUTE, TWISTED

This specification was approved on the above date by the Director of Procurement, for the use of all departments and establishments of the Government, and shall become effective not later than June 15, 1941. It may be put into effect, however, at any earlier date after promulgation.

**A. APPLICABLE SPECIFICATIONS.**

**A–1.** There are no other Federal Specifications applicable to this specification.

**B. TYPES AND GRADE.**

**B–1.** Twisted jute packing as described herein, shall be furnished in one grade and in the following types:

Type I.—Dry.
Type II.—Tarred.
Type III.—Oiled.

**C. MATERIAL AND WORKMANSHIP.**

**C–1.** See section E.

**D. GENERAL REQUIREMENTS.**

**D–1.** See section E.

**E. DETAIL REQUIREMENTS.**

**E–1.** *Kind of fiber.*—The packing shall be made of jute fiber, new or reclaimed, or from a mixture thereof. The jute fiber shall be thoroughly carded, and shall be practically free from hard, coarse fibers and extraneous matter.

**E–2.** *Type I (dry).*—

**E–2a.** *Form.*—The dry packing shall be prepared in rope form with not less than 7 nor more than 10 strands having a twist of approximately 1¼ turns per foot (1 to 2 turns per foot). The rope shall be approximately 2 inches in diameter (1½ to 2½ inches in diameter), and there shall be not less than 2½ feet nor more than 5½ feet of rope to the pound.

**E–2b.** *Limitation of oil content.*—The dry packing shall contain not more than 8 percent of oil, by total weight of the packing, as determined by the test given in par. F–2.

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E–3. *Type II (tarred).*—

E–3a. *Form.*—The tarred packing shall be prepared in rope form with not less than 7 nor more than 10 strands having a twist of approximately 1¼ turns per foot (1 to 2 turns per foot). The rope shall be approximately 1½ inches in diameter (1 to 2 inches in diameter), and there shall be not less than 1¼ feet nor more than 2¾ feet of rope to the pound. The fibers shall be thoroughly impregnated with a suitable commercial grade of pine tar.

E–4. *Type III (oiled).*—

E–4a. *Form.*—The oiled packing shall be prepared in rope form with not less than 7 nor more than 10 strands, having a twist of approximately 1¼ turns per foot (1 to 2 turns per foot). The rope shall be approximately 1½ inches in diameter (1 to 2 inches in diameter), and there shall be not less than 1¼ feet nor more than 2¾ feet of rope to the pound. The fibers shall be thoroughly impregnated with a suitable commercial grade of oil.

E–5. *Weight of shipment and tare.*—Deliveries shall be weighed under prevailing atmospheric conditions and this weight shall be recorded as gross weight. The bands and coverings shall be removed from approximately 5 percent of the number of coils in a shipment and weighed. The average of these weights shall be considered as weight of tare. (Tare in excess of 5 percent of the gross weight will not be paid for. See par. I–1.)

F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F–1. The inspector shall visually examine the material as received. A representative sample of each lot of the material shall be taken for testing.

F–2. *Determination of oil content, type I packing.*—

F–2a. *Routine test.*—A sample containing approximately 10 grams of packing shall be taken and accurately weighed. The sample shall be placed in a Soxhlet extraction apparatus and extracted with carbon tetrachloride or other suitable organic solvent. The solvent is then evaporated off and the residue dried, cooled, and weighed. The difference between the final weight and the weight of the empty flask is the weight of the soluble material, and the percentage of the weight of the original sample which the weight of soluble material represents is calculated and considered as oil content.

F–2b. *Umpire test.*—If results obtained by methods meeting the requirements of the routine test, par. F–2a, are disputed, oil content shall be determined by the following umpire test, and the results shall be considered final. A sample containing approximately 10 grams of packing shall be taken and conditioned to constant weight at 20° to 25° C. and at 50 to 65 percent relative humidity. After accurately determining the final weight, the sample shall be placed in a Soxhlet extraction apparatus and treated with ethyl ether for a period of 6 hours at an extraction rate which fills and empties the extraction tube at least 12 times per hour. The solvent extract is collected in a weighed flask and the bulk of the solvent is first distilled off and the remainder evaporated off over a steam bath. The residue is dried for 1 hour at 105° to 110° C. The flask is then cooled in a desiccator and weighed. The difference between the final weight and the weight of the empty

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flask is the weight of the soluble material, and the percentage of the weight of the original sample which the weight of soluble material represents is calculated and considered as oil content.

F–3. *Determination of weight, twist, and diameter of rope packing.*—A sample of rope packing approximately 6 feet in length, with both ends bound so as to prevent untwisting of the strands, is allowed to hang freely so as to be substantially straight. A strand is chosen and the number of complete revolutions around the axis of the rope in a definite length is counted to determine twist per foot. The approximate diameter may be obtained by measuring in at least five different places, none of which shall be within 12 inches of either end, with outside calipers and averaging. The weight per foot is determined by carefully cutting and weighing a length of 4 feet from the middle of the straightened sample.

## G. PACKING AND MARKING FOR SHIPMENT.

G–1. *Packing.*—Unless otherwise specified, the rope packing shall be prepared for shipment in commercial coils weighing approximately 50 pounds, which shall be such as to insure acceptance by common or other carriers, for safe transportation, at the lowest rate, to the point of delivery. (See par. I–2.)

G–2. *Marking.*—Unless otherwise specified, coils shall be marked by means of a tag with the name of the material, the type, and the quantity contained therein, as defined by the contract or order under which the shipment is made, the name of the contractor, and the number of the contract or order.

## H. REQUIREMENTS APPLICABLE TO INDIVIDUAL DEPARTMENTS.

H–1. *Army.*—U. S. Army Specification No. 100–2, Standard Specification for Marking Shipments. (Copies of this specification may be obtained by prospective bidders, without cost, upon application to the distributing agency indicated in the invitation for bids.)

H–2. *Navy.*—Navy Department General Specification for Inspection of Materials, copies of which may be obtained without cost upon application to the Bureau of Supplies and Accounts, Navy Department, Washington, D. C.

## I. NOTES.

I–1. Twisted jute packing should be purchased on the basis of price per pound gross weight. The gross weight should include not more than 5 percent of tare. (See par. E–5 for determination of gross weight and tare.)

I–2. Purchasers should exercise any desired options offered herein, and should state type of packing desired. The packing is prepared commercially in coils weighing approximately 50 and 100 pounds. When 100-pound coils are desired, that fact should be so specified.

I–3. The packing is intended for use in calking of bell and spigot pipe. Type I (dry) is used principally for cast iron water pipe, when joints are sealed with lead or sulphur base calking compound. Type II (tarred) is used principally for calking cast iron soil or sewer pipe and cement pipe when joints are sealed with lead or concrete. Type III (oiled) is used principally for calking vitreous pipe when joints are sealed with bituminous calking compounds.

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I–4. When characteristics of the pine tar, or mineral oil, are to be determined, the requirements applicable to this class of material should be given in the invitation for bids, either in detail or by reference to a readily available Government specification, such as Navy Department Specifications 52–T–1d; Tar, pine, or 14–0–8; Oil, mineral, light-colored (for cordage oils).

I–5. It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

I–6. This specification covers only the types, classes, grades, sizes, etc., of the commodity as generally purchased by the Federal Government, and is not intended to include all of the types, etc., which are commercially available.

**Notice.**—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

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Copies of this specification may be purchased for 5 cents.

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